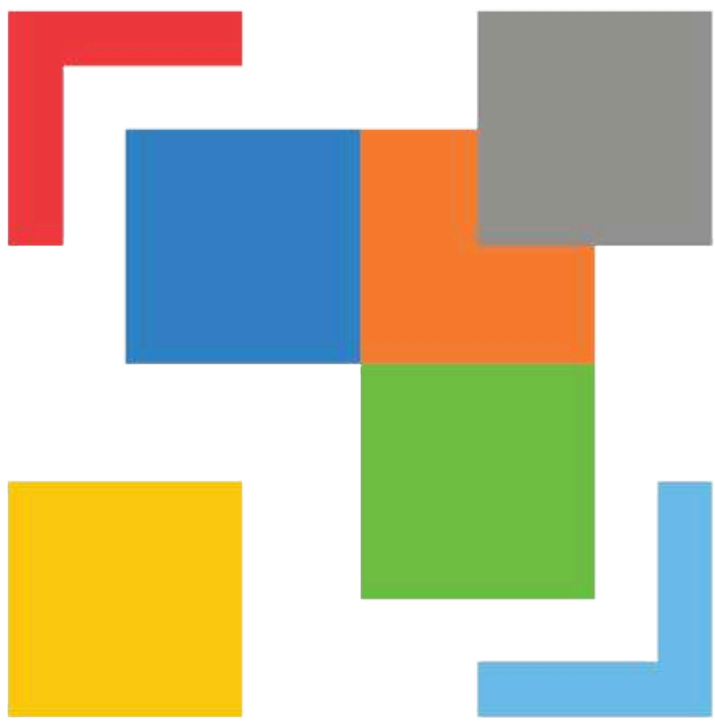




C.N.E

COMPUTER NETWORK
ENGINEERING

Choose the correct answer for questions (1-15) and complete the following table

Q(1)	Q(2)	Q(3)	Q(4)	Q(5)	Q(6)	Q(7)	Q(8)
D	D	B	D	c	e	C	a
Q(9)	Q(10)	Q(11)	Q(12)	Q(13)	Q(14)	Q(15)	
B	a	c	a	B	D	D	

سؤال
(9)

Q(1) The solution of the (separable) D. Equation $\frac{dy}{dx} = (1+y^2)(1-x^2)^{-\frac{1}{2}}$; $y\left(\frac{\sqrt{3}}{2}\right) = 0$ is

- a) $\tan^{-1} y = \sin^{-1} x + \frac{\pi}{2}$ b) $\tan^{-1} y = \sin^{-1} x - \frac{\pi}{2}$ c) $y = \sin^{-1} x - \frac{\pi}{6}$ **(d) $\tan^{-1} y = \sin^{-1} x - \frac{\pi}{3}$**

Q(2) The solution of the D. Equation $y' = (y+25x)^2$ is

- a) $\frac{1}{4} \tan^{-1}(y+4x) + x = c$ b) $\frac{1}{4} \tan^{-1}\left(\frac{y+16x}{4}\right) + x = c$
c) $\tan^{-1}\left(\frac{y+16x}{2}\right) + x = c$ **(d) $\frac{1}{5} \tan^{-1}\left(\frac{y+25x}{5}\right) + x = c$**

Q(3) The differential equation $\cos(x+y) dx + (2y+3y^2 + \cos(x+y)) dy = 0$ is

- a) Homogeneous **(b) Exact** c) Linear d) Separable

Q(4) The value of the constant c that makes $(9x^2 - cy) dx + (7x + 4y^2) dy = 0$ exact is

- a) -5 b) 5 c) -17 **(d) -7** e) none

Q(5) The integrating factor for $(3x^2y^4 + 2xy) dx + (2x^3y^3 + x^2) dy = 0$ is

- a) x^{-2} b) x^2 **(c) y^{-2}** d) x

Q(6) The substitution $u = yx^{-1}$ transforms the equation $xyy' = y^2 + x^2$ into

- a) Linear b) Homogeneous c) Separable d) Exact **(e) none**

Q(7) Every separable D. Equation is

- a) Linear b) Homogeneous **(c) Exact** d) Separable e) none

Q(8) The solution of the linear equation $x^2y' + 2xy - x + 1 = 0$, $y(1) = \frac{1}{2}$

- (a) $\frac{1}{2} - \frac{1}{x} + x^{-2}$** b) $1 - \frac{1}{x} + \frac{x^{-2}}{2}$ c) $\frac{1}{2} - \frac{1}{x} + \frac{x^{-2}}{2}$ d) $\frac{1}{x} + \frac{x^{-2}}{2}$

Q(9) The solution of the equation $\frac{dy}{dx} - 5y = -\frac{5}{2}y^3$ is

- a) $y^2 = \frac{3}{1+2ce^{-10x}}$ **(b) $y^2 = \frac{2}{1+2ce^{-10x}}$** c) $y = \frac{2}{1+2ce^{-10x}}$ d) $y^2 = \frac{2}{1+2ce^{-10x}}$

Q(10) The D.Equation $(x^2 - y^2) dx + xy dy = 0$ is

- ☒ a) Homogeneous b) Exact c) Linear d) separable e) none

Q(11) The substitution $x = u + 2, y = v + 1$ transforms $(x + 2y - 4) dx - (2x + y - 5) dy = 0$ into

- a) exact b) homogeneous ☒ c) separable d) Linear

Q(12) The substitution $u = y^{1-n}$ transforms the equation $y' + p(x)y = g(x)y^n$ into the form

- ☒ a) $u' + (1 - n)p(x)u = (1 - n)g(x)$ b) $u' + p(x)u = (1 - n)g(x)$
c) $u' + (1 - n)p(x)u = (1 - n)$ d) $u' + (1 - n)p(x)u = (1 - n)g(x)$

Q(13) The substitution $z = x + y$ transforms $(x + y + 3) dx + (3x + 3y + 4) dy = 0$ into

- a) exact ☒ b) separable c) homogeneous d) Linear

Q(14) The solution of the D.E. $y' = 6y^2x$ is

- a) $c + 3x^2$ b) $c - 3x^2$ c) $\frac{1}{3x^2}$ ☒ d) $\frac{1}{c - 3x^2}$

Q(15) The Homogeneous D.Equation is an extension to the

- a) Linear D.Equ. b) Exact D.Equ. c) Bernoulli D.Equ. ☒ d) Separable D.Equ.